Vision of the Department

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering: Dynamic learning environment that encourages problem-solving using emerging technology, · National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry–institute collaboration, Holistic, ethical and sustainable development for lifelong success.

Session 2025-2026(EVEN)

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

SHEIKH WASIMUDDIN

Name and Signature of Student and Date
(Signature and Date in Handwritten)

Shiky
6/3/26

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Session	**2024-25 (EVEN)**	**Course Name**	**Mobile and Web Application for IOT Lab**
Semester	**6**	**Course Code**	**23IOT1624**
Roll No	**63**	**Name of Student**	**SHEIKH WASIMUDDIN**

Practical Number	**05**
Course Outcome	1. Understand and apply basic Kotlin programming constructs including variables, control structures, functions, null safety, and object-oriented concepts for Android app development. 2. Design and develop responsive user interfaces using XML layout files, ViewGroups, and widgets in Android Studio, adhering to Material Design principles. 3. Implement event handling and application logic using Kotlin to capture user input, manage app state, navigate between screens, and interact with Android system components like Toasts and Dialogs 4. Integrate learned concepts to develop a mini Android application that demonstrates use of ViewModel, multi-screen navigation, and proper UI/UX design for real-world problem-solving
Aim	Understand Android Studio environment and project structure.
Problem Definition	Design UI using XML layouts and widgets. Tasks: LinearLayout, RelativeLayout, ConstraintLayout, TextView, EditText, Button.
Theory (100 words)	1. LinearLayout LinearLayout is a layout manager that organizes its child views in a single row or column. The arrangement depends on the orientation attribute, which can be set to either horizontal or vertical. Because of its simple structure, LinearLayout is widely used for creating basic interface designs where elements need to be placed one after another. 2. RelativeLayout RelativeLayout is a layout that positions interface elements in relation to other components or the parent layout. Developers can specify rules such as placing a view above, below, to the left, or to the right of another view. This layout offers more flexibility than LinearLayout and is useful for designing more structured user interfaces. 3. ConstraintLayout ConstraintLayout is an advanced layout used to design complex and responsive user interfaces in Android applications. In this layout, each UI component is positioned by defining constraints relative to other

	components or the parent container. It helps reduce deeply nested layouts and improves the overall performance of the application. 4. TextView TextView is a user interface component used to display text on the screen. It is commonly used to show information, titles, labels, or messages to users. Developers can customize properties such as text size, color, style, and alignment to improve readability. 5. EditText EditText is an input widget that allows users to enter and edit text within the application. It is commonly used in forms, search boxes, registration pages, and login screens where user input is required. Developers can also restrict the input type such as numbers, passwords, or email addresses. 6. Button Button is an interactive UI component that responds to user clicks or taps. When a user presses a button, it can trigger a specific action such as submitting information, opening a new screen, or executing a function in the application. Buttons are widely used for navigation and performing tasks in Android apps.
Procedure and Execution (100 Words)	**Step for Implementation:** 1. Open Android Studio and create a new Android project using the Empty Activity template. 2. Enter the required details such as application name, package name, and minimum SDK version and finish creating the project. 3. Navigate to the res → layout folder and open the activity_main.xml file. 4. Select or add a LinearLayout, RelativeLayout, or ConstraintLayout as the main layout container. 5. Drag and place UI widgets such as TextView, EditText, and Button from the palette into the layout. 6. Set properties for each widget such as text, hint, id, width, height, and alignment in the XML file or design view.

7. Arrange the UI elements properly using layout attributes like orientation,
constraints, or relative positioning.
8. Save the layout file and connect widgets to the activity file if required.
9. Build the project and run the application on an Android emulator or
physical device to view the designed user interface.

Code:

1. Manifest

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:tools="http://schemas.android.com/tools">

   <application
      android:allowBackup="true"
      android:dataExtractionRules="@xml/data_extraction_rules"
      android:fullBackupContent="@xml/backup_rules"
      android:icon="@mipmap/ic_launcher"
      android:label="@string/app_name"
      android:roundIcon="@mipmap/ic_launcher_round"
      android:supportsRtl="true"
      android:theme="@style/Theme.Practical5_iot63">
      <activity
         android:name=".MainActivity"
         android:exported="true">
         <intent-filter>
            <action android:name="android.intent.action.MAIN" />

            <category android:name="android.intent.category.LAUNCHER"
/>
         </intent-filter>
      </activity>
   </application>

</manifest>
```

2. MainActivity.kt

```
package com.example.practical5_iot63

import android.os.Bundle
import android.widget.*
import androidx.appcompat.app.AppCompatActivity
```

```
class MainActivity : AppCompatActivity() {

    lateinit var name: EditText
    lateinit var roll: EditText
    lateinit var dept: EditText
    lateinit var submit: Button

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        name = findViewById(R.id.etName)
        roll = findViewById(R.id.etRoll)
        dept = findViewById(R.id.etDept)
        submit = findViewById(R.id.btnSubmit)

        submit.setOnClickListener {

            val studentName = name.text.toString()
            val studentRoll = roll.text.toString()
            val studentDept = dept.text.toString()

            Toast.makeText(
                this,
                "Name: $studentName\nRoll: $studentRoll\nDept: $studentDept",
                Toast.LENGTH_LONG
            ).show()
        }
    }
}
```

3. ActivityMain.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <TextView
        android:id="@+id/title"
        android:layout_width="wrap_content"
```

```
        android:layout_height="wrap_content"
        android:text="Student Registration"
        android:textSize="24sp"
        android:textStyle="bold"
        android:layout_gravity="center"
        android:layout_marginBottom="30dp"/>

    <EditText
        android:id="@+id/etName"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Name"/>

    <EditText
        android:id="@+id/etRoll"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Roll Number"
        android:inputType="number"
        android:layout_marginTop="15dp"/>

    <EditText
        android:id="@+id/etDept"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Department"
        android:layout_marginTop="15dp"/>

    <Button
        android:id="@+id/btnSubmit"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Submit"
        android:layout_marginTop="25dp"/>

</LinearLayout>
```

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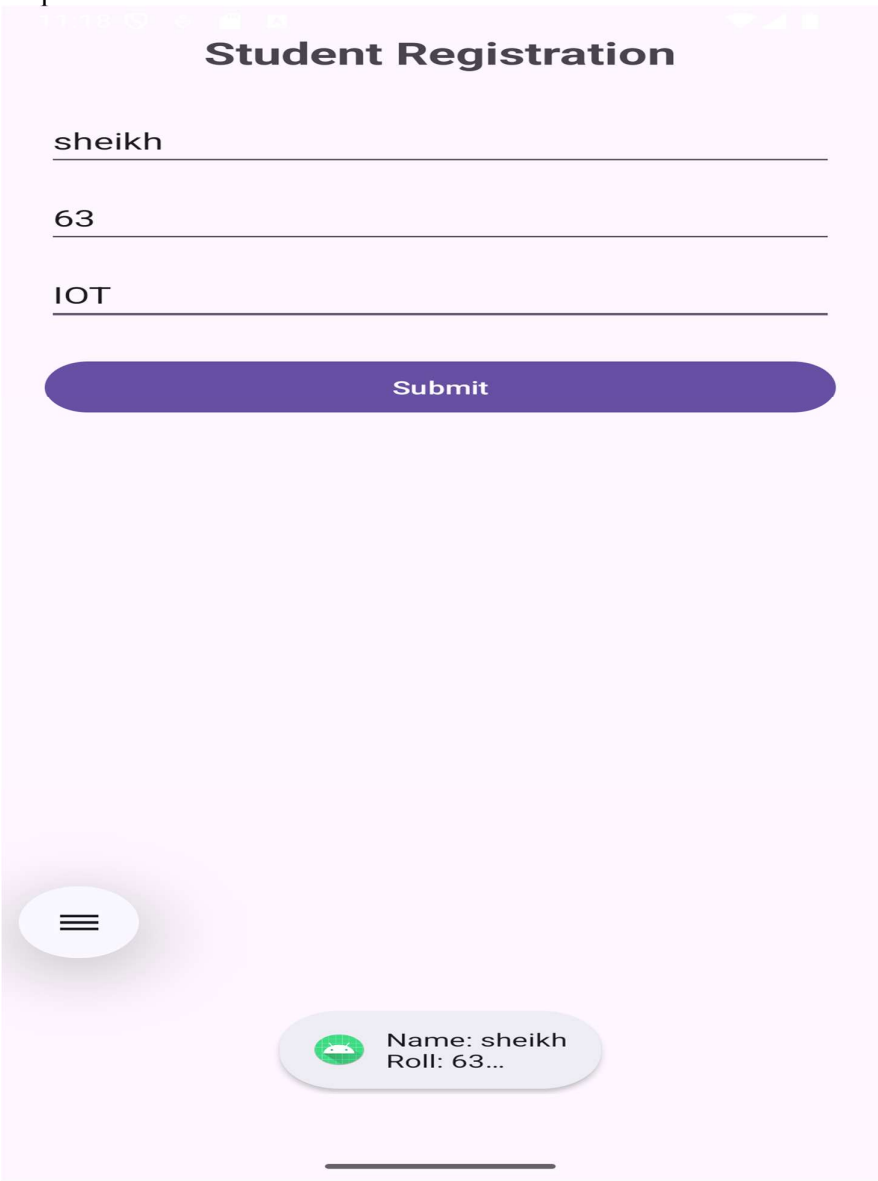
Output:



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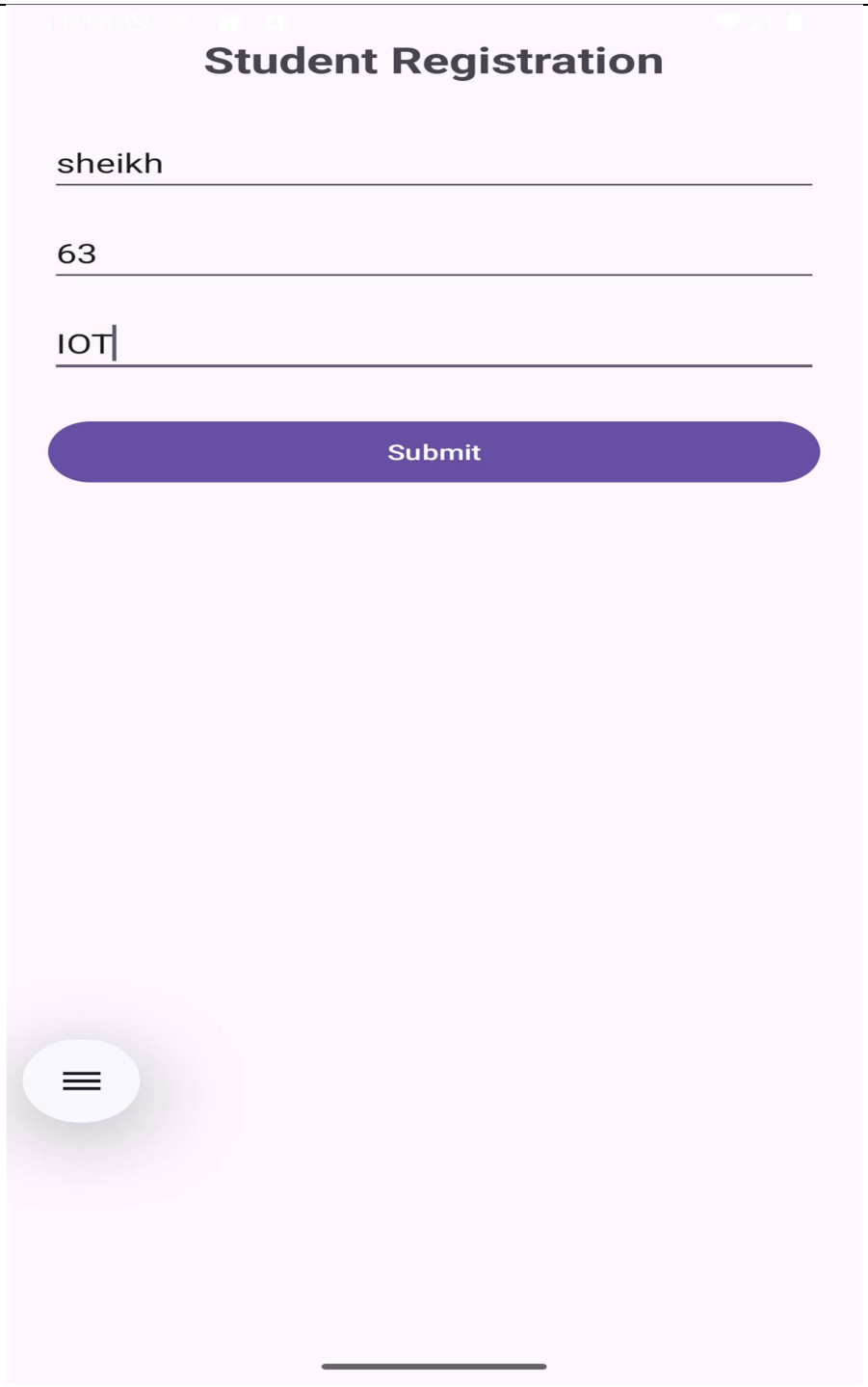

Output Analysis	After implementing the XML layout, the user interface is displayed correctly when the application runs on the emulator or Android device. The selected layout type such as LinearLayout, RelativeLayout, or ConstraintLayout arranges the UI components according to the defined properties. The TextView displays text information, the EditText allows users to enter input, and the Button appears as a clickable element for performing actions. The alignment, spacing, and positioning of the widgets follow the layout attributes specified in the XML file. This confirms that the UI design using XML layouts and widgets is functioning properly and the interface is structured as intended.
Link of student Github profile where lab assignment has been uploaded	https://github.com/sheikhwasimuddin
Conclusion	From this experiment, the process of designing a user interface in Android using XML layouts and widgets was successfully understood. Different layout types such as LinearLayout, RelativeLayout, and ConstraintLayout were used to arrange UI components effectively. Widgets like TextView, EditText, and Button were implemented to display information, accept user input, and perform actions. This experiment helped in understanding how Android layouts control the structure and appearance of an application screen. Overall, it provided practical knowledge about creating and organizing user interface elements using XML in Android Studio.

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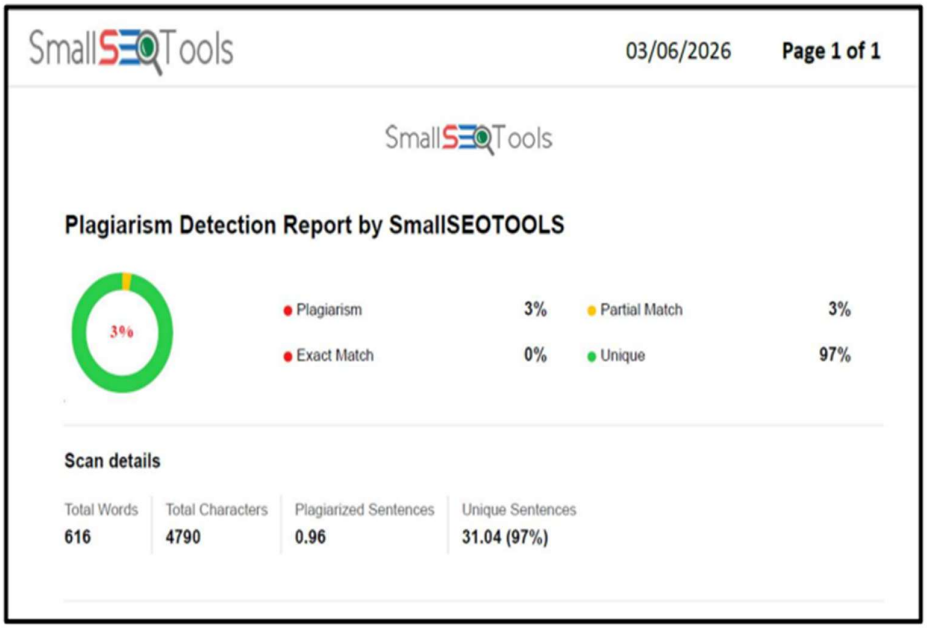
Plag Report (Similarity index < 12%)	
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